EXERCISE SOLUTIONS TO *THE RISING SEA: FOUNDATIONS OF ALGEBRAIC GEOMETRY*

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Contents

1. Just enough category theory to be dangerous

1.1. Categories and functors.

Exercise (1.1.A)**.** *Groupoids.*

(a) *Make sense of the statement "a group is a groupoid with one element."*

Solution. Let $\mathcal{G}$ be a one element groupoid, call the single element x. Then by definition the identity morphism is in $\hom_{\mathcal{G}}(x, x)$. We will take this to be the identity element of the group $\hom_{\mathcal{G}}(x, x)$. Any two morphisms are composable, which gives the group composition, and by definition of groupoid any morphism is invertible, yielding the group inverse.

Conversely, if we have a group G we can make a category with one object whose morphisms are the elements of G, and whose identity, composition, and inverses are given by the group structure. This yields a groupoid. □

(b) *Give an example of a groupoid that is not a group.*

Solution. A category with more than one object, and only the identity morphisms. □

Exercise (1.1.B)**.** *If A is an object in a category $\mathcal{C}$, show that the invertible elements of* $\hom_{\mathcal{C}}(A, A)$ *form a group, called the automorphism group. What are the automorphism groups in* Set *and* Vect_k*?*

Proof. This is exactly the same as part 1 above. The identity morphism serves as the identity, the inverse morphisms yield the group inverses, and the category axioms yield composition.

The automorphism groups in Set are the symmetric groups, and the automorphism groups in Vect_k are the general linear groups. □

Exercise (1.1.C.)**.** *Fix a field k. Let $(-)^{\vee\vee}$ be the double dual functor from the category of finite-dimensional k-vector spaces to itself. Show that $(-)^{\vee\vee}$ is naturally isomorphic to the identity functor.*

Proof. For a fixed vector space V, define the map $\alpha_V : V \to V^{\vee\vee}$ by $v \mapsto (\phi \mapsto \phi(v))$. This is injective because if $\phi \mapsto \phi(v)$ is the zero map for every ϕ, then $v = 0$, so by finite-dimensionality α_V is an isomorphism. And it remains to check that for any map of vector spaces $f : V \to W$ and $v \in V$, that

$$\begin{aligned}\alpha_W(f(v)) &= (\phi \mapsto \phi(f(v))) \\ &= (\psi \mapsto \psi(v)) \circ f^\vee \\ &= f^{\vee\vee}(\psi \mapsto \psi(v)) \\ &= f^{\vee\vee}(\alpha_V(v))\end{aligned}$$

which completes the proof. □

Exercise (1.1.D). *Let $\mathcal{V}$ be the category whose objects are the k-vector spaces k^n for $n \geq 0$, and whose morphisms are linear transformations. There is a functor F from $\mathcal{V}$ to the category of f.d. k-vector spaces sending each vector space to itself. Show that F is an equivalence of categories.*

Proof. Following Vakil's remark about ignoring set-theoretic conditions, choose a basis for each finite dimensional k-vector space. For any vector space of the form k^n, choose the standard basis $\{e_i\}$.

Then define a functor G from f.d. k-vector spaces to $\mathcal{V}$ by sending an n-dimensional k-vector space to k^n. For a morphism ϕ from V to an m-dimensional vector space W, let $\{v_1, \ldots, v_n\}$ be the chosen basis for V and $\{w_1, \ldots, w_m\}$ the chosen basis for W. Then define $G\phi$ to be the matrix whose (i, j)th entry is the coefficient of w_j in $\phi(v_i)$.

Define a natural transformation α from FG to the identity functor where $\alpha_V : FGV \to V$ sends $FGV = k^n$ to V by mapping the standard basis vector e_i of FGV to the chosen ith basis vector v_i of V. Then α_V is an isomorphism. For naturality, we have for a map $\phi : V \to W$ that

$$\begin{aligned}\phi(\alpha_V(e_i)) &= \phi(v_i) \\ &= \sum_j (G\phi)_{i,j} w_j \\ &= \alpha_W\Big(\sum_j (G\phi)_{i,j} e_j\Big) \\ &= \alpha_W(FG\phi)(e_i).\end{aligned}$$

In the other direction, GF is the identity functor, which completes the proof. □

1.2. Universal properties determine an object up to unique isomorphism.

Exercise (1.2.A). *Show that any two initial objects are uniquely isomorphic, and any two final objects are uniquely isomorphic.*

Proof. Let i_1, i_2 be initial. Then there is exactly one map $f_{12} : i_1 \to i_2$ and likewise $f_{21} : i_2 \to i_1$. Since there is also exactly one map from i_1 to itself, the identity, so because $f_{21} \circ f_{12}$ is one such map, it must be the identity. Likewise $f_{12} \circ f_{21}$ is the identity, which yields the unique isomorphisms. The argument is exactly the same for final objects. □

Exercise (1.2.B). *What are the initial and final objects (if they exist) in* Ring, Set, *and* Top, *or the category of subsets of a set under inclusion, or the category of open subsets of a topological space under inclusion?*

Solution. In Ring, the initial object is the integers, since 1 mapping to 1 dictates where the remaining integers go; the terminal object is the zero ring, since everything must map to zero. In Set, the initial object is the empty set; the final object is the singleton. In Top, the initial object is the empty space; the final object is the singleton. In the category of subsets or open sets, the initial object is the empty set/space; the terminal object is the entire set/space. □

Exercise (1.2.C)**.** *Let A be a commutative ring and S a multiplicative subset of A. Show that $f : A \mapsto S^{-1}A$ via $a \mapsto a/1$ is injective if and only if S contains no zerodivisors.*

Proof. First suppose f is not injective, i.e. there is a nonzero element $a \in A$ such that $a/1 = 0/1$. So there exists some $s \in S$ such that $s(a1 - 01) = sa = 0$. Then s is a zerodivisor.

Conversely, suppose S contains a zerodivisor s, with $st = 0$ for t nonzero. Then $0 = st = s(t1 - 01)$, $t/1 = 0/1 = 0$, so $f(t) = t/1 = 0/1$ and f fails to be injective. ☐

Exercise (1.2.D)**.** *Verify that $\iota : A \to S^{-1}A$ is initial among A-algebras B where every element of S is sent to an invertible element in B, i.e. any map $\phi : A \to B$ sending all elements of S to invertible elements in B factors through $S^{-1}A$.*

Proof. Define $\tilde{\phi} : S^{-1}A \to B$ via $\tilde{\phi}(a/s) := \phi(a)\phi(s)^{-1}$. This makes sense since we have assumed that $\phi(s)$ is invertible. Let $a, b \in A$ and $s, t \in S$. Then

$$\tilde{\phi}(ab/st) = \phi(ab)\phi(st)^{-1} = \phi(a)\phi(b)\phi(s)^{-1}\phi(t)^{-1} = \tilde{\phi}(a/s)\tilde{\phi}(b/t)$$

and

$$\begin{aligned}\tilde{\phi}(a/s + b/t) &= \tilde{\phi}((at + sb)/st)\\ &= \phi(at + sb)\phi(st)^{-1}\\ &= \phi(a)\phi(t)/\phi(s)\phi(t) + \phi(s)\phi(b)/\phi(s)\phi(t)\\ &= \tilde{\phi}(a/s) + \tilde{\phi}(b/t).\end{aligned}$$

so $\tilde{\phi}$ is a ring homomorphism. And $\tilde{\phi}(\iota(a)) = \tilde{\phi}(a/1) = \phi(a)/\phi(1) = \phi(a)$ as desired. ☐

Exercise (1.2.E)**.** *The localization of an A-module M defined via the universal property, actually exists.*

Proof. Following the hint, define $S^{-1}M$ to have elements of the form m/s where $m \in M$ and $s \in S$, subject to the equivalence relation where $m/s \sim m'/t$ if there exists an element $s' \in S$ such that $s'(ms - m't) = 0$, with the standard operation and $S^{-1}A$-module operation. Define $\iota : M \to S^{-1}M$ via $m \mapsto m/1$.

Let $\phi : M \to N$ be an A-module map such that multiplication by any element of s is an isomorphism $N \to N$. Then we can define a map $\tilde{\phi} : S^{-1}M \to N$ via $m/s \mapsto s^{-1}\phi(m)$, which makes sense taking s^{-1} to be the inverse to multiplication by s. This is an A-module map and $\tilde{\phi}(\iota(m)) = \tilde{\phi}(a/1) = 1^{-1}\phi(a) = \phi(a)$. ☐

Exercise (1.2.F)**.** *Properties of localization.*

(a) *Localization commutes with finite products, i.e. for A-modules $M_1, \dots, M_n$ provide an A-module isomorphism $S^{-1}(M_1 \times \cdots M_n) \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$.*

Proof. Define our map via $(m_1, \dots, m_n)/s \mapsto (m_1/s, \dots, m_n/s)$. For surjectivity, given an element $(m_1/s_1, \dots, m_n/s_n)$, the element

$$(s_2 \cdots s_n m_1, \dots, s_1 \cdots s_{n-1} m_n)/s_1 \cdots s_n$$

will map to it. For injectivity, suppose $(m_1/s, \dots, m_n/s)$ is zero, i.e. there exist elements $t_i \in S$ such that $t_i m_i = 0$ for all i. Then $t_1 \cdots t_n(m_1, \dots, m_n) = 0$, so $(m_1, \dots, m_n)/s = 0$. ☐

(b) *Localization commutes with arbitrary direct sums.*

Proof. Let Λ be an indexing set. Define a map $S^{-1}(\bigoplus_{\lambda \in \Lambda} M_\lambda) \to \bigoplus_{\lambda \in \Lambda} S^{-1}M_\lambda$ via

$$(m_\lambda)_{\lambda \in \Lambda}/s \mapsto (m_\lambda/s)_{\lambda \in \Lambda}.$$

For surjectivity, given an element $(m_\lambda/s_\lambda)_{\lambda \in \Lambda}$, we know that only finitely many indices are nonzero, say $\lambda_1, \dots, \lambda_k$. Then let $s = \lambda_1 \cdots \lambda_k$, and consider the element $(k_\lambda m_\lambda)_{\lambda \in \Lambda}$ where

$k_{\lambda_i} = s/s_i$ for the nonzero indices and zero otherwise. Then it maps to what we want. For injectivity, suppose $(m_\lambda/s)_{\lambda\in\Lambda}$ is zero, i.e. there exist elements t_λ such that $t_\lambda m_\lambda = 0$ for all λ. Again only finitely many of the m_λ are nonzero, so take the corresponding t_λ and let their product be t. Then $t(m_\lambda)_{\lambda\in\Lambda} = 0$ and we are done. □

(c) *Localization does not commute with infinite products, i.e. the same map defined as above is not always an isomorphism for infinite products.*

Proof. We know that $\mathbf{Z}$ localized at $\mathbf{Z} \setminus \{0\}$ is $\mathbf{Q}$, so that $\prod_{\mathbf{N}} \mathbf{Q}$ contains for example the element $x := (1, 1/2, 1/3, \ldots)$. On the other hand, suppose there were an element in $(\mathbf{Z} \setminus \{0\})^{-1}(\mathbf{Z}^{\mathbf{N}})$ mapping to x, i.e. some element $(a_1, a_2, \ldots)/s$ where $a_i \in \mathbf{Z}$ and $s \in \mathbf{Z} \setminus \{0\}$ such that $a_n/s = 1/n$ for all $n \in \mathbf{N}$. Then $na_n = s$ for all $n \in \mathbf{N}$, i.e. s is an integer divisible by all natural numbers, a contradiction. □

Exercise (1.2.G)**.** *Show that* $\mathbf{Z}/10 \otimes \mathbf{Z}/12 \cong \mathbf{Z}/2$.

Proof. We define a map $\tilde{f} : \mathbf{Z}/10 \times \mathbf{Z}/12 \to \mathbf{Z}/2$ via $([a], [b]) \mapsto [ab]$. This is clearly bilinear. If $a = a' + 10n$ and $b = b' + 12m$, so $[ab] = [a'b' + 12ma' + 10nb' + 120nm] = [a'b']$ in $\mathbf{Z}/2$, so this map is well defined. By the universal property of tensor products, f factors through the tensor product to yield a map $f : \mathbf{Z}/10 \otimes \mathbf{Z}/12 \to \mathbf{Z}/2$ such that $[a] \otimes [b] \mapsto [ab]$.

Consider the map $g : \mathbf{Z}/2 \to \mathbf{Z}/10 \otimes \mathbf{Z}/12$ given by $[a] \mapsto [a] \otimes [1]$. This is again bilinear, and if $a = a' + 2n$, then $[a] \otimes [1] = [a' + 2n] \otimes [1] = [a'] \otimes [1] + [12n - 10n] \otimes [1] = [a'] \otimes [1] + [1] \otimes [12n] - [10n] \otimes [1] = [a']$, so this is well defined.

We moreover have $fg([a]) = f([a] \otimes [1]) = [a]$ and $gf([a], [b]) = g([ab]) = [ab] \otimes [1] = [a] \otimes [b]$. Hence f is an isomorphism. (Note that this argument works to show in general that $\mathbf{Z}/m \otimes \mathbf{Z}/n \cong \mathbf{Z}/\gcd(m, n)$). □

Exercise (1.2.H)**.** *Show that* $(-) \otimes_A N$ *is a covariant functor* $\mathsf{Mod}_A \to \mathsf{Mod}_A$*, and that it is right exact.*

Proof. Let $f : M \to M'$ be an A-module map. Then $(m, n) \to f(m) \otimes n$ is bilinear, so it yields a map $f_* : M \otimes N \to M' \otimes N$, which on generators is given by $m \otimes n \mapsto f(m) \otimes n$. This respects composition, thus making $(-) \otimes N$ into a functor.

Let $M \xrightarrow{f} M' \xrightarrow{g} M'' \xrightarrow{h} 0$ be exact. Tensoring, we get a complex

$$M \otimes N \xrightarrow{f_*} M' \otimes N \xrightarrow{g_*} M'' \otimes N \xrightarrow{h_*} 0,$$

which we need to check is exact. We know by the first isomorphism theorem that g induces an isomorphism $M'/\operatorname{im}(f) \to M''$.

Claim 1.1. Tensor products commute with quotients.

Proof. Let P, Q, R be A-modules. Suppose Q is a submodule of P. Consider the map $\tilde{f} : (P/Q) \times R \to (P \otimes R)/(Q \otimes R)$ given by $([p], r) \mapsto [p \otimes r]$. If $[p] = [p']$, then $[p \otimes r] - [p' \otimes r] = [p - p' \otimes r] = [q \otimes r] = 0$ so this map is well defined. One can also check bilinearity. Hence it extends to a map $f : (P/Q) \otimes R \to (P \otimes R)/(Q \otimes R)$.

We now define a map $\tilde{g} : P \times R \to (P/Q) \otimes R$ via $(p, r) \mapsto [p] \otimes r$. This is bilinear, hence yields a map $g' : P \otimes R \to (P/Q) \otimes R$, which further factors through the quotient by $Q \otimes R$, yielding a map $g : (P \otimes R)/(Q \otimes R) \to (P/Q) \otimes R$. And f, g are mutually inverse, yielding the claim. □

By the claim, g_* induces an isomorphism (since functors send isos to isos)

$$M'' \otimes N \leftarrow (M'/\operatorname{im}(f)) \otimes N \cong (M' \otimes N)/\operatorname{im}(f_*)$$

so $\operatorname{im}(f_*) = \ker(g_*)$. And surjectivity of g_* follows directly from surjectivity of g. □

Exercise (1.2.I)**.** *The tensor product (T, t) of two modules M, N is unique up to unique isomorphism. (Here $t : M \times N \to T$ is the bilinear map given by the universal property.)*

Proof. Let (T', t') be another pair satisfying the universal property. Then t' must factor through T, i.e. there exists a unique map $f : T \to T'$ such that $f \circ t = t'$. Likewise, there exists a unique map $g : T' \to T$ such that $g \circ t' = t$. Plugging these two formulas into each other we get $f \circ g \circ t' = t'$ and $g \circ f \circ t = t$.

Now we use the universal property again. Since t' is a bilinear map from $M \times N$ to T', it must factor uniquely through T'. One such factoring is as above, $(f \circ g) \circ t' = t'$, but another factoring is $\mathrm{id}_{T'} \circ t'$, so by uniqueness, $f \circ g = \mathrm{id}_{T'}$. Likewise, $g \circ f = \mathrm{id}_T$, hence f, g are inverses, and the unique isomorphisms between T, T'. □

Exercise (1.2.J)**.** *Show that the tensor product actually exists, i.e. the construction from §1.2.5 satisfies the universal property.*

Proof. Recall that $M \otimes N$ is the free A-module generated by $M \times N$, quotiented by the following relations:

(a) $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$
(b) $(m, n_1 + n_2) - (m, n_1) - (m, n_2)$
(c) $a(m, n) - (am, n)$
(d) $a(m, n) - (m, an)$

for $a \in A, m, m_1, m_2 \in M, n, n_1, n_2 \in N$. Let $f : M \times N \to P$ be a bilinear A-module morphism. It follows by bilinearity that

(a) $f((m_1 + m_2, n) - (m_1, n) - (m_2, n)) = f(m_1, n) + f(m_2, n) - f(m_1, n) - f(m_2, n) = 0$
(b) $f((m, n_1 + n_2) - (m, n_1) - (m, n_2)) = f(m, n_1) + f(m, n_2) - f(m, n_1) - f(m, n_2) = 0$
(c) $f(a(m, n) - (am, n)) = af(m, n) - af(m, n) = 0$
(d) $f(a(m, n) - (m, an)) = af(m, n) - af(m, n) = 0$

so f factors uniquely through the quotient. □

Exercise (1.2.K)**.** *"Important exercise"*

(a) *If M is an A-module and $\phi : A \to B$ is a ring morphism, give $B \otimes_A M$ the structure of a B-module. Show that this gives a functor $\mathsf{Mod}_A \to \mathsf{Mod}_B$.*

Proof. Regard B as an A-module via the action $A \times B \to B : (a, b) \mapsto \phi(a)b$. Now $B \otimes_A M$ makes sense as an A-module. Now for fixed $b \in B$, consider the map $\tilde{f}_b : B \times M \to B \otimes_A M$ given by $(b', m) \to bb' \otimes m$. We compute

$$\tilde{f}_b(\phi(a)b', m) = b\phi(a)b' \otimes m = \phi(a)(bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

and

$$\tilde{f}_b(b', am) = bb' \otimes am = a \cdot (bb' \otimes m) = a \cdot \tilde{f}_b(b', m)$$

so $\tilde{f}_b$ is A-bilinear, and it induces a map $f_b : B \otimes_A M \to B \otimes_A M$, which yields the B-module structure on $B \otimes_A M$.

To see that this is a functor, suppose we have an A-module morphism $g : M \to N$. Then we get a map $B \times M \to B \otimes_A N$ given by $(b, m) \mapsto b \otimes g(m)$. This is again A-bilinear, so it induces a map $g_* : B \otimes_A M \to B \otimes_A N$, which acts on elementary tensors via $b \otimes m \mapsto b \otimes g(m)$. □

(b) *Following notation above, if $\psi : A \to C$ is a ring morphism, then $B \otimes_A C$ has a ring structure.*

Proof. Again, C becomes an A-module via the action $(a, c) \mapsto \psi(a)c$. Then $B \otimes_A C$ automatically gets an abelian group structure. To define the ring product, note that specifying a map

$$(B \otimes C) \times (B \otimes C) \to B \otimes C$$

is the same thing as specifying a map

$$B \otimes C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C)$$

so we can start here. First define a map

$$B \times C \to \hom_{\mathsf{Mod}_A}(B \times C, B \otimes C) : (b_1, c_1) \mapsto ((b_2, c_2) \mapsto b_1 b_2 \otimes c_1 c_2).$$

By properties of the tensor product, the image of this map is an A-bilinear map, so we get a map

$$B \times C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C) : b_1, c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2).$$

Again, this is also A-bilinear. For example we can check

$$\phi(a) b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto \phi(a) b_1 b_2 \otimes c_1 c_2) = (b_2 \otimes c_2 \mapsto \phi(a)(b_1 b_2 \otimes c_1 c_2)).$$

Hence we get a map

$$B \otimes C \to \hom_{\mathsf{Mod}_A}(B \otimes C, B \otimes C) : b_1 \otimes c_1 \mapsto (b_2 \otimes c_2 \mapsto b_1 b_2 \otimes c_1 c_2)$$

which is equivalent to the map

$$\mu : (B \otimes C) \times (B \otimes C) \to B \otimes C : (b_1 \otimes c_1), (b_2 \otimes c_2) \mapsto b_1 b_2 \otimes c_1 c_2$$

which is A-bilinear by definition since it comes from the universal property. Now for $x, y, z \in B \otimes C$, we have by bilinearity

$$\mu(x + y, z) = \mu(x, z) + \mu(y, z), \quad \mu(x, y + z) = \mu(x, y) + \mu(x, z)$$

so this satisfies distributivity. The identity element is $1_B \otimes 1_C$, and μ is obviously associative, so we are done. □

Exercise (1.2.L)**.** *If S is a multiplicative subset of A and M is an A-module, describe a natural isomorphism $(S^{-1}A) \otimes_A M \to S^{-1}M$ (as both A-modules and $S^{-1}A$-modules).*

Proof. Consider first the map $S^{-1}A \times M \to S^{-1}M$ given by $(a/s, m) \mapsto (a \cdot m)/s$. This is A-bilinear since

$$(a' \cdot a/s, m) = (a'a/s, m) \mapsto (aa' \cdot m)/s = a \cdot (a' \cdot m)/s$$

and

$$(a/s, a' \cdot m) \mapsto (a \cdot (a' \cdot m))/s = (aa' \cdot m)/s = a \cdot (a' \cdot m)/s.$$

Hence this induces a map $f : (S^{-1}A) \otimes_A M \to S^{-1}M$ which acts on elementary tensors as

$$f(a/s \otimes m) = (a \cdot m)/s.$$

We can define an inverse map as follows. First consider the map $\tilde{g} : M \to (S^{-1}A) \otimes_A M$ via $m \mapsto 1/1 \otimes m$. Since multiplication by elements of S are invertible in $(S^{-1}A) \otimes_A M$, the map $\tilde{g}$ factors through the localization to yield a map $g : S^{-1}M \to (S^{-1}A) \otimes_A M$ which acts via $m/s \mapsto s^{-1}(1/1 \otimes m) = (1/s \otimes m)$. Then we can check

$$gf(a/s \otimes m) = g((a \cdot m)/s) = (1/s \otimes a \cdot m) = (a/s \otimes m)$$

and

$$fg(m/s) = f(1/s \otimes m) = m/s.$$

It remains to check that f is also a $S^{-1}A$-module map. Recall from Exercise 1.2.K.1 that $S^{-1}A \otimes_A M$ has the structure of a $S^{-1}A$-module via the map $A \to S^{-1}A, a \mapsto a/1$. So for $a/s, a'/s' \in S^{-1}A, m \in M$, we have

$$f(a/s \cdot (a'/s') \otimes m) = f(aa'/ss' \otimes m) = (aa' \cdot m)/ss' = (a'/s') \cdot (am/s)$$

as desired. For naturality, we need to check that the following diagram commutes for any map of A-modules $\phi : M \to N$:

$$\begin{tikzcd}
(S^{-1}A) \otimes_A M \arrow[r, "f"] \arrow[d, "\phi_*"'] & S^{-1}M \arrow[d, "\phi_*"] \\
(S^{-1}A) \otimes_A N \arrow[r, "f"'] & S^{-1}N
\end{tikzcd}$$

but this is not too bad, for following the top and right arrows we have

$$\phi_*(f(a/s \otimes m)) = \phi_*((a \cdot m)/s) = \phi(a \cdot m)/s$$

and following the left and bottom ones we have

$$f(\phi_*(a/s \otimes m)) = f(a/s \otimes \phi(m)) = (a \cdot \phi(m))/s = \phi(a \cdot m)/s$$

as desired. □

Exercise (1.2.M)**.** *Show that tensor products commute with arbitrary direct sums, i.e. for A-modules M and* $\{N_\alpha\}_{\alpha \in \Lambda}$*, define an isomorphism* $M \otimes (\oplus_{\alpha\in\Lambda} N_\alpha) \to \oplus_{\alpha \in \Lambda}(M \otimes N_\alpha)$*.*

Proof. As with every map out of a tensor prouduct, we begin by defining a map

$$\tilde{f} : M \times (\oplus_{\alpha\in\Lambda} N_\alpha) \to \oplus_{\alpha\in\Lambda}(M \otimes N_\alpha), (m, (n_\alpha)_{\alpha\in\Lambda}) \mapsto (m \otimes n_\alpha)_{\alpha\in\Lambda}.$$

This, being bilinear, induces a map $f : M \otimes (\oplus_{\alpha\in\Lambda} N_\alpha) \to \oplus_{\alpha\in\Lambda}(M \otimes N_\alpha)$ which acts on elementary tensors via $m \otimes (n_\alpha)_{\alpha\in\Lambda} \mapsto (m \otimes n_\alpha)_{\alpha\in\Lambda}$.

For each $\alpha \in \Lambda$, consider the map

$$\tilde{g}_\alpha : M \times N_\alpha \to M \otimes (\oplus_{\alpha\in\Lambda} N_\alpha), (m, n) \mapsto m \otimes \iota_\alpha(n)$$

where $\iota_\alpha : N_\alpha \to \oplus_\alpha N_\alpha$ is the inclusion. This is A-bilinear, so it induces a map $g_\alpha : M \otimes N_\alpha \to M \otimes (\oplus_{\alpha\in\Lambda} N_\alpha)$. The sum of all the g_α yields a map $g := \oplus_{\alpha\in\Lambda}(M \otimes N_\alpha) \to M \otimes (\oplus_{\alpha\in\Lambda} N_\alpha)$.

And we check

$$\begin{aligned} fg((m_\alpha \otimes n_\alpha)_{\alpha\in\Lambda}) &= f \sum_{\alpha\in\Lambda} (m_\alpha \otimes \iota_\alpha(n_\alpha)) \\ &= \sum_{\alpha\in\Lambda} f(m_\alpha \otimes \iota(n_\alpha)) \\ &= \sum_{\alpha\in\Lambda} (\ldots, 0, m_\alpha \otimes n_\alpha, 0, \ldots) \\ &= (m_\alpha \otimes n_\alpha)_{\alpha\in\Lambda} \end{aligned}$$

and

$$gf(m \otimes (n_\alpha)_{\alpha\in\Lambda}) = g((m \otimes n_\alpha)_{\alpha\in\Lambda}) = \sum_{\alpha\in\Lambda} m \otimes \iota_\alpha(n_\alpha) = m \otimes \sum_{\alpha\in\Lambda} \iota_\alpha(n_\alpha) = m \otimes (n_\alpha)_{\alpha\in\Lambda}$$

so f is an isomorphism. □

Exercise (1.2.H)**.** *Show that in* Set*, the fiber product is given by* $X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}$*.*

Proof. Call the right hand side of above R. Let W be a set with maps $f_X : W \to X$ and $f_Y : W \to Y$ such that $\alpha \circ f_X = \beta \circ f_Y$. Define the map $\phi : W \to R$ via $\phi(w) = (f_X(w), f_Y(w))$. This actually lands in R since $\alpha \circ \pi_X(\phi(w)) = \alpha \circ f_X(w) = \beta \circ f_Y(w) = \beta \circ \pi_Y(\phi(w))$. Then clearly $\pi_X \circ \phi = f_X$, and $\pi_Y \circ \phi = f_Y$. Thus R satisfies the universal property so it is *the* fiber product in the category of sets. □

Exercise (1.2.O). *If X is a topological space, show that fiber products always exist in the category of open sets of X by describing what a fibered product is.*

Proof. Remember that the morphisms here are just inclusions of open sets. So suppose we have open sets U, U' mapping into W, i.e. $U, U' \subset W$. We claim that the fiber product here is just $U \cap U'$ (intersections of two open sets are open), with the morphisms given by inclusions into U, U' . To see this, given another open set V with morphisms $V \to U$ and $V \to U'$ that agree with $U \to W$ and $U' \to W$, then we know that V must be a subset of both U and U'. It follows that V is a subset of $U \cap U'$. And by definition, all the morphisms are inclusions, so everything commutes. □

Exercise (1.2.P). *If Z is the final object in the category $\mathcal{C}$, show that the fiber product $X \times_Z Y$ is given by the product $X \times Y$. (Assume all (fiber) products exist.)*

Proof. Let $W \in \mathcal{C}$ with maps $f_X : W \to X$ and $f_Y : W \to Y$ as in the setup of the universal property. Then by the universal property of products, we get a map $\phi : W \to X \times Y$ which factors f_X and f_Y. Moreover, since Z is the final object, the compositions $X \times Y \to X \to Z$ and $X \times Y \to Y \to Z$ must agree. □

Exercise (1.2.Q). *Stacking Cartesian squares gives a Cartesian rectangle:*

$$\begin{tikzcd}
U \arrow[r] \arrow[d] & V \arrow[d] \\
W \arrow[r] \arrow[d] & X \arrow[d] \\
Y \arrow[r] & Z
\end{tikzcd}$$

Proof. Suppose we have maps $f_Y : A \to Y$ and $f_V : A \to V$ such that their postcompositions with $Y \to Z$ and $V \to Z$ agree. Then since the bottom square is Cartesian, there is a unique map $f_{AW} : A \to W$ which factors f_Y and $f_{VX} \circ f_V$, i.e. $f_{WX} \circ f_{AW} = f_{VX} \circ f_V$. Now since the top square is Cartesian, there is a unique map $f_{AW} : A \to U$ factoring f_{AW} (hence f_Y) and f_V, completing the proof. See below for a figure of all the maps involved:

$$\begin{tikzcd}
A \arrow[drr, bend left, "f_V"] \arrow[dr, dashed, "f_{AU}"] \arrow[ddr, dashed, "f_{AW}"] \arrow[dddr, bend right, "f_Y"'] & & \\
& U \arrow[r] \arrow[d] & V \arrow[d, "f_{VX}"] \\
& W \arrow[r, "f_{WX}"] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z
\end{tikzcd}$$

□

Exercise (1.2.R). *Given morphisms $X_1 \to Y$, $X_2 \to Y$, and $Y \to Z$, there is a natural morphism $X_1 \times_Y X_2 \to X_1 \times_Z X_2$ (assuming both fiber products exist).*

Proof. Consider the following diagram:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[dr, dashed] \arrow[drrr, bend left] \arrow[dddr, bend right] & & & \\
& X_1 \times_Z X_2 \arrow[rr] \arrow[dd] & & X_2 \arrow[d] \\
& & & Y \arrow[d] \\
& X_1 \arrow[r] & Y \arrow[r] & Z
\end{tikzcd}$$

where the outside arrows commute by definition. Hence by the universal property we get the dotted morphism. □

Exercise (1.2.S)**.** *Show that the diagonal-base-change diagram below is Cartesian:*

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\
Y \arrow[r] & Y \times_Z Y
\end{tikzcd}$$

Proof. The top arrow is defined by the diagram from the previous exercise. The left arrow is either composition $X_1 \times_Y X_2 \to X_i \to Y$. The right arrow is the dotted arrow below:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r] \arrow[d] \arrow[dr, dashed] & X_2 \arrow[dr] & \\
X_1 \arrow[dr] & Y \times_Z Y \arrow[r] \arrow[d] & Y \arrow[d] \\
& Y \arrow[r] & Y
\end{tikzcd}$$

And the bottom arrow is the diagonal map. Stacking the top and right arrow diagrams in front of each other, padding the middle, we get a diagram as follows:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[dr, dashed, red] \arrow[drrrr, bend left, blue] \arrow[ddddr, bend right, blue] & & & & & & \\
& X_1 \times_Z X_2 \arrow[rrr] \arrow[dr, red, "\mathrm{id}"] \arrow[ddd] & & & X_2 \arrow[dr, blue, "\mathrm{id}"] \arrow[ddd] & & \\
& & X_1 \times_Z X_2 \arrow[rrr] \arrow[dr, dashed, red] \arrow[ddd] & & & X_2 \arrow[dr, blue] & \\
& & & Y \times_Z Y \arrow[rrr] \arrow[ddd] & & & Y \arrow[ddd] \\
& X_1 \arrow[rrr] \arrow[dr, blue, "\mathrm{id}"] & & & Z \arrow[ddrr] & & \\
& & X_1 \arrow[dr, blue] & & & & \\
& & & Y \arrow[rrr] & & & Z
\end{tikzcd}$$

So the composition of the red arrows is the unique map coming from the universal property applied to the front square and composition of the blue arrows. On the other hand, if we stack the left and bottom arrow diagrams on top of each other, padding the middle, we get

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[rr, bend left, blue] \arrow[dr, dashed, red] \arrow[dd, bend right, blue] & & X_2 \arrow[drr, bend left, blue] & & & & \\
& Y \times_Z Y \arrow[rrr] \arrow[ddd] \arrow[dr, red] & & & Y \arrow[dr, blue] \arrow[ddd] & & \\
X_1 \arrow[ddr, bend right, blue] & & Y \arrow[rrr] \arrow[ddd] \arrow[dr, dashed, red] & & & Y \arrow[dr, blue] & \\
& & & Y \times_Z Y \arrow[rrr] \arrow[ddd] & & & Y \arrow[ddd] \\
& Y \arrow[rrr] \arrow[dr, blue] & & & Z \arrow[ddrr] & & \\
& & Y \arrow[dr, blue] & & & & \\
& & & Y \arrow[rrr] & & & Z
\end{tikzcd}$$

But now the blue arrows on the outside are the same, so by the universal property, the composition of the red arrows in both diagrams must agree, so the square in the exercise commutes.

Now let A be an object with morphisms to Y and $X_1 \times_Z X_2$ such that their postcompositions with the according maps are equal, i.e. the following diagram commutes:

$$\begin{tikzcd}
A \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\
& Y \arrow[r] & Y \times_Z Y
\end{tikzcd}$$

There being a morphism $A \to X_1 \times_Z X_2$ implies that we have morphisms $f_i : A \to X_i$ for $i = 1, 2$, whose postcompositions with $X_i \to Y \to Z$ are equal. The right arrow above is induced by maps $X_1 \times_Z X_2 \to X_i \to Y$, which both have to agree with $A \to Y$, that is they are equal. Hence the compositions $A \to X_i \to Y$ induce a map $A \to X_1 \times_Y X_2$, completing the proof. $\square$

Exercise (1.2.T)**.** *Show that the coproduct in the category of sets is disjoint union.*

Proof. Let X, Y be sets and let us take the disjoint union to be

$$X \coprod Y := \{(x, 0) : x \in X\} \cup \{(y, 1) : y \in Y\}.$$

Define the inclusions $\iota_X : X \to X \coprod Y$ via $x \mapsto (x, 0)$ and $\iota_Y : Y \to X \coprod Y$ via $y \mapsto (y, 1)$. Let A be a set with maps $f_X : X \to A$, $f_Y : Y \to A$. Then we can define a map $f : X \coprod Y \to A$ via $(x, 0) \mapsto f_X(x)$ and $(y, 1) \mapsto f_Y(y)$, which makes everything commute. For uniqueness, if $g : X \coprod Y \to A$ is another map making things commute, we have

$$g(x, 0) = g(\iota_X(x)) = f_X(x) = f(x, 0)$$

and likewise for Y. $\square$

Exercise (1.2.U). *For ring morphisms $A \to B$, $A \to C$, show that there is a natural morphism $B \to B \otimes_A C$ given by $b \mapsto b \otimes 1$ and likewise $C \to B \otimes_A C$ given by $c \mapsto 1 \otimes c$. Show that these maps make the following a cocartesian diagram in the category of rings.*

$$\begin{tikzcd} A \arrow[r, "f_B"] \arrow[d, "f_C"'] & B \arrow[d] \\ C \arrow[r] & B \otimes_A C \end{tikzcd}$$

Proof. To see that $b \to b \otimes 1$ is a ring map, we have $b + b' \mapsto (b + b') \otimes 1 = b \otimes 1 + b' \otimes 1$, and $bb' \mapsto bb' \otimes 1 = (b \otimes 1)(b' \otimes 1)$, and $1 \mapsto 1 \otimes 1$. Likewise for $C \to B \otimes_A C$. The square commutes since $f_B(a) \otimes 1 = f_B(a) \cdot 1 \otimes 1 = 1 \otimes f_C(a) \cdot 1 = 1 \otimes f_C(a)$.

Let D be another ring with maps $g_B : B \to D$ and $g_C : C \to D$ such that $g_B f_b = g_C f_C$. Define a map $B \otimes_A C \to D$ beginning with $B \times C \to D, (b, c) \mapsto g_B(b) g_C(c)$. By virtue of g_B being a ring morphism, we have

$$(f_B(a)b, c) \mapsto g_B(f_B(a)b) g_C(c) = g_B(f_B(a)) g_B(b) g_C(c)$$

and likewise in the second argument, so this map is A-bilinear. Consequently we have a map $\phi : B \otimes_A C \to D$ acting on basis elements via $b \otimes c \mapsto g_B(b) g_C(c)$. Then $g_B(b) = g_B(b)1 = g_B(b) g_C(1) = \phi(b \otimes 1)$, and likewise $g_C(c) = \phi(1 \otimes c)$, so we are done. ☐

Exercise (1.2.V). *Show that the composition of two monomorphisms is a monomorphism.*

Proof. Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be monomorphisms. Suppose $\pi_1, \pi_2 : W \to X$ satisfy $(g \circ f) \circ \pi_1 = (g \circ f) \circ \pi_2$. Then by associativity of composition, we have $g \circ (f \circ \pi_1) = g \circ (f \circ \pi_2)$, and because g is a mono, $f \circ \pi_1 = f \circ \pi_2$, and because f is a mono, $\pi_1 = \pi_2$. ☐

Exercise (1.2.W). *Show that a morphism $\pi : X \to Y$ is a monomorphism if and only if the fiber product $X \times_Y X$ exists and the induced diagonal map $\delta_\pi : X \to X \times_Y X$ is an isomorphism.*

Proof. Forward direction. We claim that the fiber product is given by any object X' with a chosen isomorphism $\psi : X' \to X$ serving as both projection maps. This exists, for example, by taking $X' = X$ and $\psi = \text{id}$. To see this is the fiber product, it is clear that the square commutes. Given any maps $f_1, f_2 : W \to X$ such that $\pi \circ f_1 = \pi \circ f_2$, we know that $f_1 = f_2$ since π is a monomorphism. Then defining $W \to X$ by $\psi^{-1} \circ f_1$ makes things commute. If (X'', ψ'') is another fiber product, then $\psi''^{-1} \circ \psi$ gives the unique isomorphism, so (X', ψ) satisfies the universal property. Finally, the diagonal map must satisfy $\psi \circ \delta_\pi = \text{id}$, so it is an isomorphism since ψ is an isomorphism.

Backward direction. By assumption, δ_π is an isomorphism, and it satisfies the identity $\text{pr}_X \circ \delta_\pi = \text{id}$, so both projection maps pr_X are inverse to δ_π, so both projection maps are the same. Then given maps $f_1, f_2 : W \to X$ such that $\pi \circ f_1 = \pi \circ f_2$, by the the universal property of fiber products we get a unique map $\phi : W \to X \times_Y X$ such that $f_1 = \delta_\pi^{-1} \circ \phi = f_2$. ☐

Exercise (1.2.X). *Given morphisms $X_1 \to Y$, $X_2 \to Y$, and a monomorphism $Y \to Z$, show that $X_1 \times_Y X_2 \to X_1 \times_Z X_2$ is an isomorphism.*

Proof. By the previous exercise, g in the diagonal-base-change diagram below is an isomorphism

$$\begin{tikzcd} X_1 \times_Y X_2 \arrow[r, "f"] \arrow[d, "h"'] & X_1 \times_Z X_2 \arrow[d, "j"] \\ Y \arrow[r, "g"'] & Y \times_Z Y \end{tikzcd}$$

We can proceed by showing the more general fact that in an arbitrary Cartesian square

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "h"'] & B \arrow[d, "j"] \\ C \arrow[r, "g"] & D \end{tikzcd}$$

that if g is an isomorphism then so is f. The trick is to first consider the square

$$\begin{tikzcd} B \arrow[r, "\mathrm{id}"] \arrow[d, "g^{-1}\circ j"'] & B \arrow[d, "j"] \\ C \arrow[r, "g"] & D \end{tikzcd}$$

which we claim is Cartesian. It obviously commutes. Given maps $\phi_B : X \to B$ and $\phi_C : X \to C$ such that $j \circ \phi_B = g \circ \phi_C$, the map we want $X \to B$ is just given by ϕ_B, since $\phi_B = \mathrm{id} \circ \phi_B$, and $g^{-1} \circ j \circ \phi_B = g^{-1} \circ g \circ \phi_C = \phi_C$. So B is another fiber product, and by the universal property there is an isomorphism $u : A \to B$ making the following diagram commute

$$\begin{tikzcd} A \arrow[drr, bend left, "f"] \arrow[dr, dashed, "u"] \arrow[ddr, bend right, "h"'] & & \\ & B \arrow[r, "\mathrm{id}"'] \arrow[d, "g^{-1}\circ j"'] & B \arrow[d, "h"] \\ & C \arrow[r, "g"'] & D \end{tikzcd}$$

Because f also makes the diagram commute, $u = f$ and we are done. □

Exercise (1.2.Y)**.** *Yoneda lemma.*

(a) *Let A, A' be objects in a category $\mathcal{C}$. Suppose there exist a collection of maps $i_C : \mathrm{Hom}(C, A) \to \mathrm{Hom}(C, A')$ for $C \in \mathcal{C}$ that commute with the maps $\mathrm{Hom}(Z, X) \to \mathrm{Hom}(Y, X)$ induced by morphisms $Y \to Z$. (i.e. a natural transformation between the functors $\mathrm{Hom}(-, A)$ and $\mathrm{Hom}(-, A')$). Show that the collection of maps $\{i_C\}$ are induced by a unique morphism $g : A \to A'$, i.e. each i_C is of the form $u \mapsto g \circ u$.*

Proof. Consider plugging in $C = A$. Then we have a map $i_A : \mathrm{Hom}(A, A) \to \mathrm{Hom}(A, A')$. We know that $\mathrm{id}_A \in \mathrm{Hom}(A, A)$, and we claim that $g = i_A(\mathrm{id}_A)$. Now suppose we have a morphism $\phi : C \to A$. By the naturality/commutativity assumption, this gives rise to a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(A, A) \arrow[r, "i_A"] \arrow[d, "-\circ\phi"'] & \mathrm{Hom}(A, A') \arrow[d, "-\circ\phi"] \\ \mathrm{Hom}(C, A) \arrow[r, "i_C"'] & \mathrm{Hom}(C, A') \end{tikzcd}$$

Tracing through the element id_A, the top-right path ends up at $i_A(\mathrm{id}_A) \circ \phi = g \circ \phi$, while the bottom-left path ends up at $i_C(\mathrm{id}_A \circ \phi) = i_C(\phi)$. It follows that $i_C(\phi) = g \circ \phi$, so we are done. □

(b) *If furthermore the i_C are all bijections, show that g is an isomorphism. (So if we have a natural isomorphism of the functors $\mathrm{Hom}(-, A)$ and $\mathrm{Hom}(-, A')$ then we also get an isomorphism $A \to A'$.)*

Proof. Since $i_{A'}$ is an isomorphism, there is some $f \in \mathrm{Hom}(A', A)$ such that $i_{A'}(f) = \mathrm{id}_{A'}$. Now consider the same commutative diagram as above with $C = A'$. Let the vertical arrows be precomposition by f. Then commutativity yields $g \circ f = i_A(\mathrm{id}_A) \circ f = \mathrm{id}_{A'}$.

On the other hand, flip this commutative diagram diagonally and swap the direction of the horizontal arrows to get

$$\begin{tikzcd} \mathrm{Hom}(A', A') \arrow[r, "i_{A'}^{-1}"] \arrow[d, "- \circ g"'] & \mathrm{Hom}(A', A) \arrow[d, "- \circ g"'] \\ \mathrm{Hom}(A, A') \arrow[r, "i_A^{-1}"] & \mathrm{Hom}(A, A) \end{tikzcd}$$

which yields $f \circ g$ following the top and right paths and id_A following the bottom left. So g is an isomorphism. □

Exercise (1.2.Z)**.** *Yoneda lemma again.*

(a) Let A, B be objects in a category $\mathcal{C}$. Give a bijection between $\mathrm{Hom}(\mathrm{Hom}(A, -), \mathrm{Hom}(B, -))$ *and* $\mathrm{Hom}(B, A)$.

Solution. (This is Exercise 1.2.Z.c, take $F = \mathrm{Hom}(B, -)$.) For the exact same proof, read on. Given a natural transformation $\alpha \in \mathrm{Hom}(\mathrm{Hom}(A, -), \mathrm{Hom}(B, -))$, define $\Phi(\alpha) = \alpha_A(\mathrm{id}_A) \in \mathrm{Hom}(B, A)$. Conversely, given $g \in \mathrm{Hom}(B, A)$, define a natural transformation by $\Psi(g)_C(f) = f \circ g$. This is indeed natural since given any morphism $\varphi : C \to C'$, the square

$$\begin{tikzcd} \mathrm{Hom}(A, C) \arrow[r, "- \circ g"] \arrow[d, "\varphi \circ -"'] & \mathrm{Hom}(B, C) \arrow[d, "\varphi \circ -"'] \\ \mathrm{Hom}(A, C') \arrow[r, "- \circ g"'] & \mathrm{Hom}(B, C') \end{tikzcd}$$

commutes by associativity of morphism composition.

Now we need to check that Φ and Ψ are inverses. In one direction, given a natural transformation α, $C \in \mathcal{C}$, and $f \in \mathrm{Hom}(A, C)$, we have

$$(\Psi \circ \Phi(\alpha))_C(f) = (\Psi(\alpha_A(\mathrm{id}_A)))_C(f) = f \circ \alpha_A(\mathrm{id}_A).$$

By naturality of α, the square

$$\begin{tikzcd} \mathrm{Hom}(A, A) \arrow[r, "\alpha_A"] \arrow[d, "f \circ -"'] & \mathrm{Hom}(B, A) \arrow[d, "f \circ -"] \\ \mathrm{Hom}(A, C) \arrow[r, "\alpha_C"'] & \mathrm{Hom}(B, C) \end{tikzcd}$$

yields the equality $f \circ \alpha_A(\mathrm{id}_A) = \alpha_C(f)$, which shows $\Psi \circ \Phi$ is the identity.

In the other direction, let $g \in \mathrm{Hom}(B, A)$. We compute

$$\Phi(\Psi(g)) = \Psi(g)_A(\mathrm{id}_A) = \mathrm{id}_A \circ g = g$$

which completes the proof. □

(b) State and prove the dual statement for $\mathrm{Hom}(\mathrm{Hom}(-, A), \mathrm{Hom}(-, B))$ *and* $\mathrm{Hom}(A, B)$.

Proof. Let $\Phi : \mathrm{Hom}(\mathrm{Hom}(-, A), \mathrm{Hom}(-, B)) \to \mathrm{Hom}(A, B)$ be the map $\alpha \mapsto \alpha_A(\mathrm{id}_A)$ from Exercise 1.2.Y.a. Conversely given $g \in \mathrm{Hom}(A, B)$ define a natural transformation $\Psi(g)$ with $\Psi(g)_C(f) = g \circ f$ where $f \in \mathrm{Hom}(C, A)$. Then, given a natural transformation α, $C \in \mathcal{C}$, and $f \in \mathrm{Hom}(C, A)$, we have

$$\Psi(\Phi(\alpha))_C(f) = \Psi(\alpha_A(\mathrm{id}_A))_C(f) = \alpha_A(\mathrm{id}_A) \circ f.$$

Again, by naturality we get a commutative diagram

$$\begin{tikzcd} \mathrm{Hom}(A,A) \arrow[r, "\alpha_A"] \arrow[d, "-\circ f"'] & \mathrm{Hom}(A,B) \arrow[d, "-\circ f"] \\ \mathrm{Hom}(C,A) \arrow[r, "\alpha_C"'] & \mathrm{Hom}(C,B) \end{tikzcd}$$

which yields the equality $\alpha_A(\mathrm{id}_A) \circ f = \alpha_C(f)$, so $\Psi(\Phi(\alpha)) = \alpha$. In the other direction, for $g \in \mathrm{Hom}(A, B)$, we have

$$\Phi(\Psi(g)) = (\Psi(g))_A(\mathrm{id}_A) = g \circ \mathrm{id}_A = g$$

which completes the proof!!!!!! □

(c) *Let* $F : \mathcal{C} \to \mathsf{Set}$ *be a covariant functor. Give a bijection between* $\mathrm{Hom}(\mathrm{Hom}(A, -), F)$ *and* FA.

Solution. We do the exact same thing as before. In one direction, given a natural transformation $\alpha : \mathrm{Hom}(A, -) \Rightarrow F$, define $\Phi(\alpha)$ to be $\alpha_A(\mathrm{id}_A)$. In the other direction, given $X \in FA$, $C \in \mathcal{C}$ and $f \in \mathrm{Hom}(A, C)$, define $\Psi(X)_C(f) = (Ff)(X)$.

We check, again, that given a natural transformation $\alpha : \mathrm{Hom}(A, -) \Rightarrow F$, $C \in \mathcal{C}$ and $f \in \mathrm{Hom}(A, C)$, that

$$\Psi(\Phi(\alpha))_C(f) = \Psi(\alpha_A(\mathrm{id}_A))_C(f) = (Ff)(\alpha_A(\mathrm{id}(A)))$$

which by naturality is equal to $\alpha_C(f)$. And on the other hand, given an element $X \in FA$, we have

$$\Phi(\Psi(X)) = \Psi(X)_A(\mathrm{id}_A) = (F\,\mathrm{id}_A)(X) = \mathrm{id}\,FA(X) = X$$

which completes the proof. □